

**Sardar Vallabhbhai National Institute of Technology (SVNIT),
Surat Department of Computer Science and Engineering
B.Tech II Semester-III
CS231 – Object Oriented Programming Practical Assignment
Lab Assignment - 6**

1. Write a programme to implement a Smart Home Management System using the C++ concepts of Static members, Nested member functions and using objects as function arguments. Design a system to manage rooms and smart devices, focusing on tracking shared resources and demonstrating object relationships. Your designed system should implement the following requirements,
 - a. The House management system contains multiple Room objects.
 - b. Each Room contains multiple device objects.
 - c. Track and report on shared resources (Smart Devices)
 - d. **Device Management:** The system must be able to add devices to specific rooms.
 - e. **Power Consumption:** Calculate and report the total power consumption for the entire house by summing up the consumption of all devices.
 - f. **Summary:** Your program must display a clear, organised list of all rooms and the devices contained within each, along with their individual power consumption.
2. Create a class BankAccount where each object stores balance, and a static data member stores interestRate. Write a static function to change the interest rate for all accounts..
3. Create a class BankAccount with static member interestRate. Create an array of

objects for multiple customers. Write a function that accepts an object as an argument and calculate interest.

4. Create a programme using a pointer to navigate and manipulate the data in an array of integers.
 - a. Declare an array with 10 integers.
 - b. Declare a pointer and point it to the first element of an array.
 - c. Access and print the value of the fifth element in the array without using array indexing ([]). (**Pointer Arithmetic**)
 - d. Declare another pointer to an integer and point it to the integer pointer created in the previous steps, and print the value of the element it points to. (**Pointer to Pointer**)
 - e. Update the value of the third element using the first pointer created in step b, and print the value of the third element using the second pointer created in step d.
5. Develop a C++ program to manage the installation of CCTV cameras for a security company, where the number of cameras required for each project varies. Use dynamic memory allocation to handle the data efficiently for storing information about CCTV cameras.
 - a. Implement the programme in C-style (using *malloc()/calloc()/free()*).
 - i. Create a Camera class with the following data members: *model*, *status* and *installation_area*.
 - ii. Prompt the user for the number of cameras and Camera details.
 - iii. Dynamically allocate the memory (using an appropriate function or operator) for an array of Camera objects.

- iv. Create a function to display the details.
 - v. Finally, deallocate the memory
 - b. Implement the previous programme using C++ operators (*new* and *delete*)
 - c. Expand the previous C-style programme to add additional CCTV cameras using *realloc()* to efficiently resize the memory. Prompt for an additional number of cameras and it's details from the user.
- 6. Write a program for a College Library System using a Book class with data members: bookID, title, author, and copies.
 - a. Define both a default constructor and a parameterised constructor.
 - b. Add a const member function display() to show book details.
 - c. Write a friend function that takes two Book objects, compares their number of copies, and returns the book with more copies.
 - d. In main(), create at least two books, use the friend function, and display the details of the book with more copies.
- 7. Write a program to store and manage student marks dynamically using malloc, calloc, realloc, and free. The program should:
 - a. Input the number of students and allocate memory using malloc.
 - b. Input and display the marks.
 - c. Allocate memory for another set of students using realloc and input additional marks.
 - d. Use calloc to initialize a pass/fail array.
 - e. Display all marks and pass/fail status.

f. Deallocate all memory using free.

8. Hospital Management System (Patient Records)

Create a system where the hospital can store patient records dynamically. Each patient has:

- Name
- Age
- Disease
- ID number

Allow the user to:

- Add new patients dynamically
- Display all patient records
- Delete a patient by ID

9. Book Management Using calloc()

Create a book management system where each book has:

Title

Author

Price

Steps:

Allocate memory for n books using calloc()

Input book data

Display all books

Free memory

10. Dynamic Array Average Calculator

Use calloc() to allocate memory for n floating-point numbers

Calculate and print the average

Free memory after use

